CSE 6179 – Digital Forensics

Hands-on Lab # 5 **Digital Forensics using Autopsy** **Spring 2026**

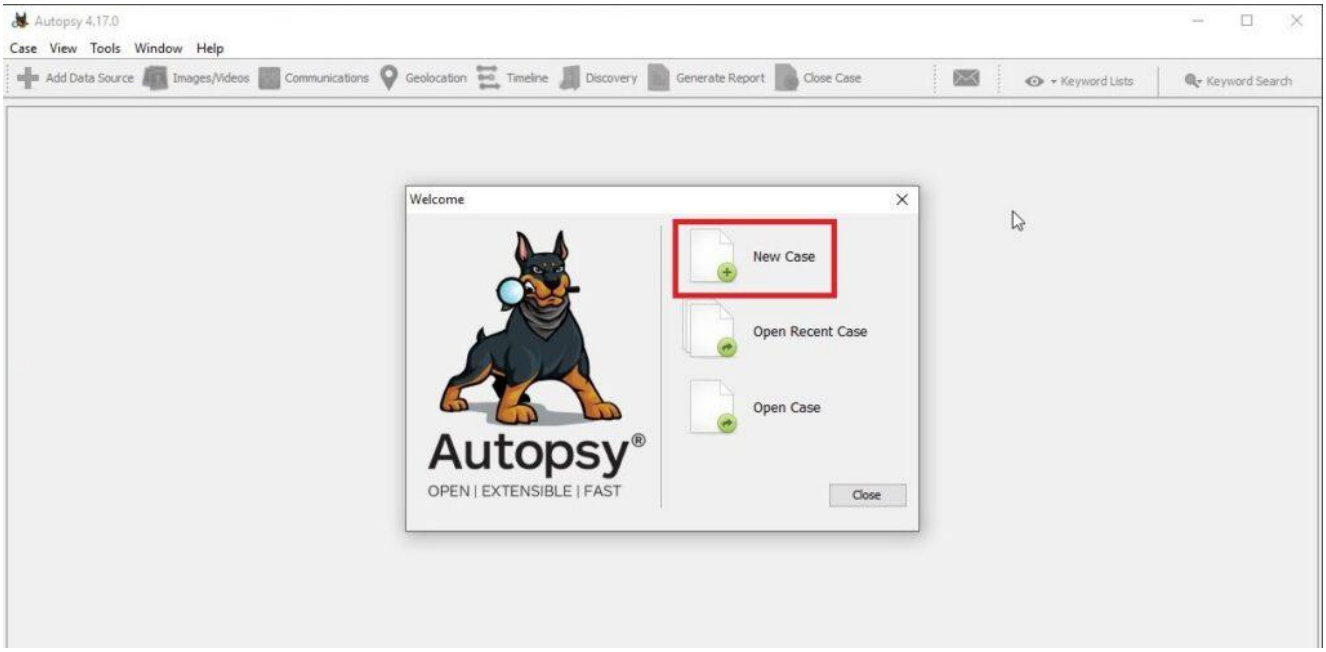
For this lab, we will use an Encase Image of a **suspected Dell Latitude laptop** named "**Hacking Case**" that can be downloaded from the following link:

https://cfreds.nist.gov/all/NIST/HackingCase

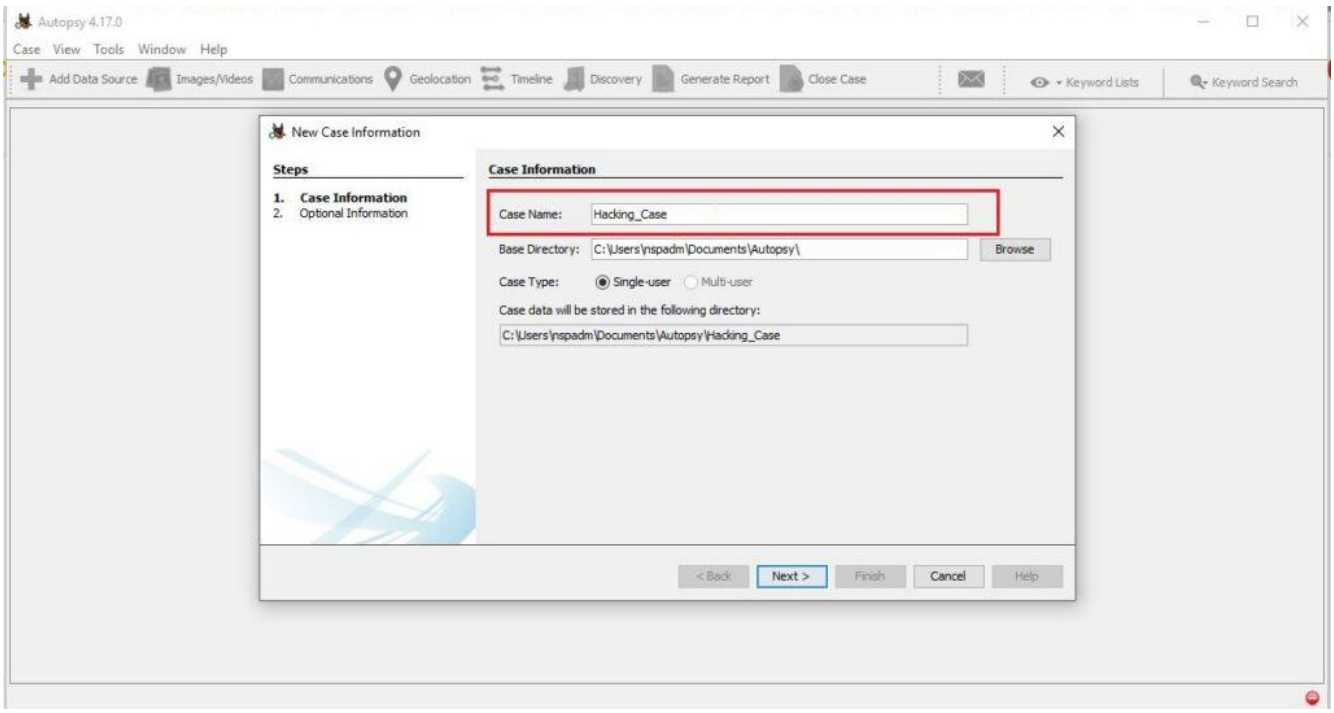
Here is the **background scenario** about this image.

"On 09/20/04, a Dell CPi notebook computer, serial # VLQLW, was found abandoned along with a wireless PCMCIA card and an external homemade 802.11b antennae. It is suspected that this computer was used for hacking purposes, although cannot be tied to a hacking suspect, G=r=e=g S=c=h=a=r=d=t. (The equal signs are just to prevent web crawlers from indexing this name; there are no equal signs in the image files.) Schardt also goes by the online nickname of "Mr. Evil" and some of his associates have said that he would park his vehicle within range of Wireless Access Points (like Starbucks and other T-Mobile Hotspots) where he would then intercept internet traffic, attempting to get credit card numbers, usernames & passwords. Find any hacking software, evidence of their use, and any data that might have been generated. Attempt to tie the computer to the suspect, G=r=e=g S=c=h=a=r=d=t. A DD image and a EnCase image of the abandoned computer have already been made."

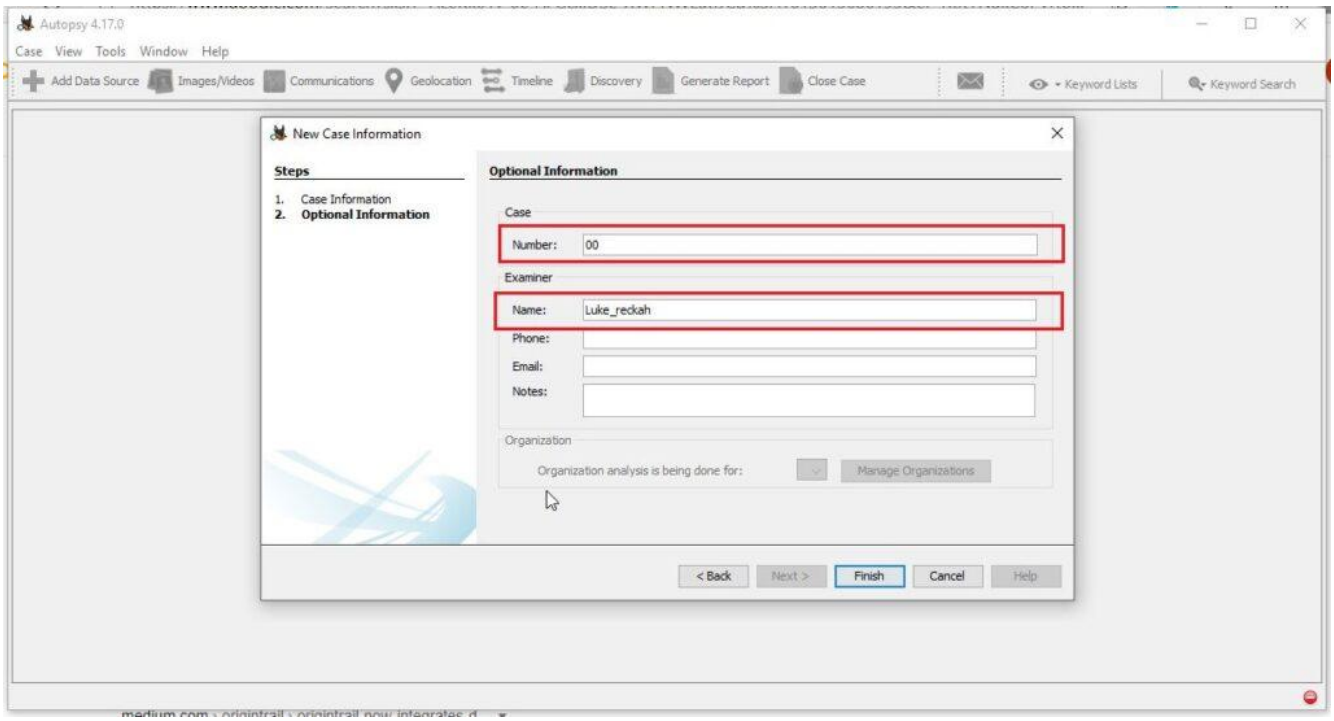
The mission for us is to analyze this Encase Image and answer around **19 questions** that solve this case. The questions are also provided by the same people who provided this Hacking Case to us. Let's start analyzing this image and solve the case. Once the program is installed, open it and click on "**New Case**".



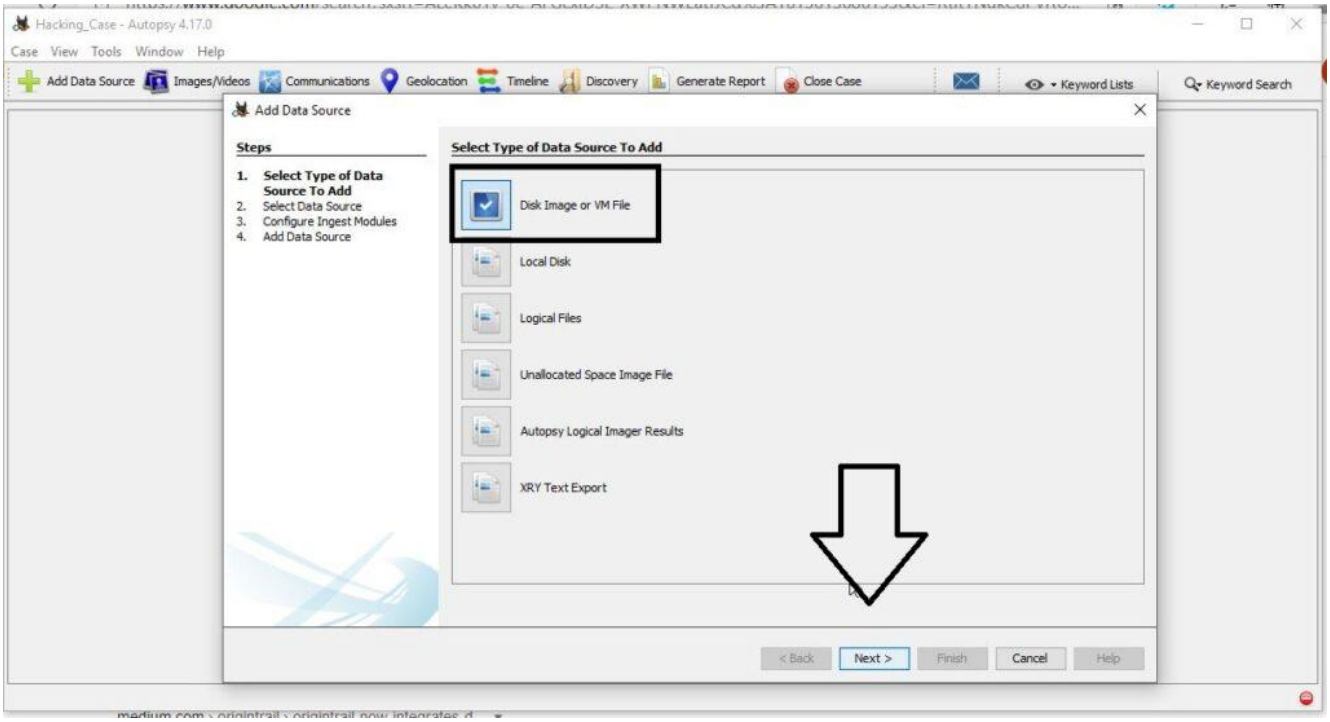
Give a name to the case. We have named it "**Hacking_Case**".



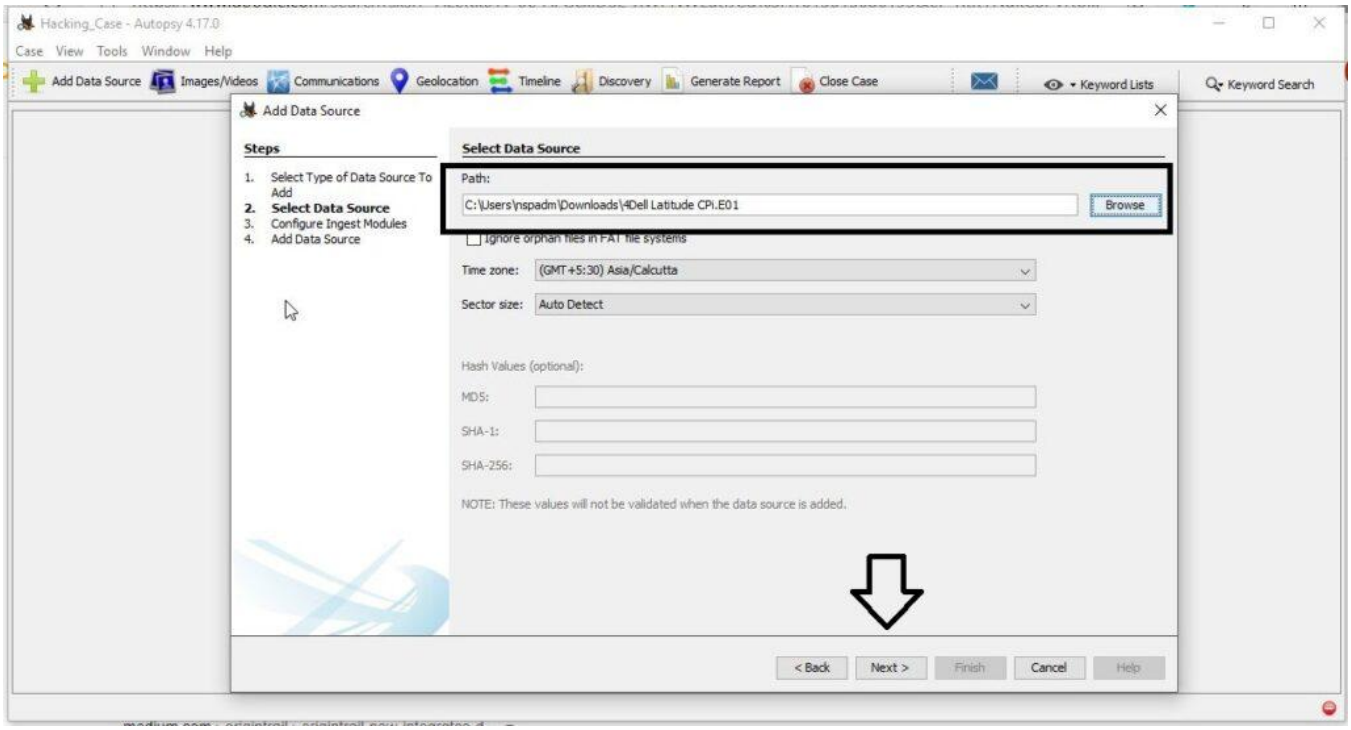
Assign a number to the case and provide the name of the Forensic investigator. Our case number is 00 and the investigator is Luke Reckah.



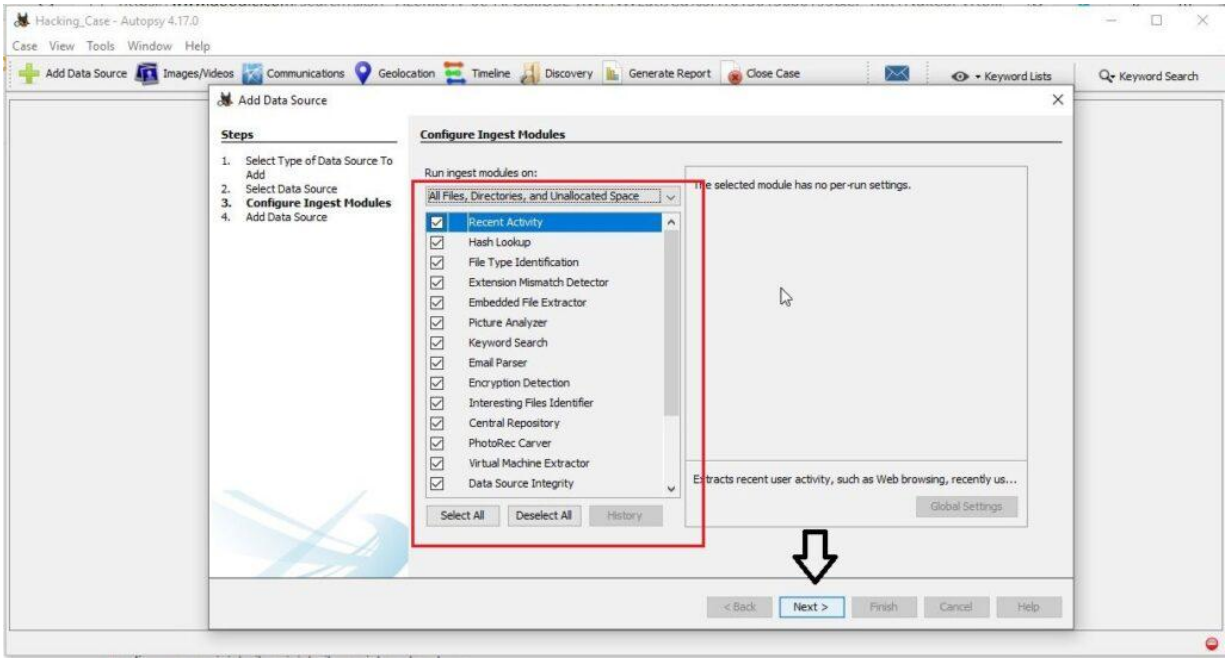
Next, select the type of source. Select "**Disk Image**".

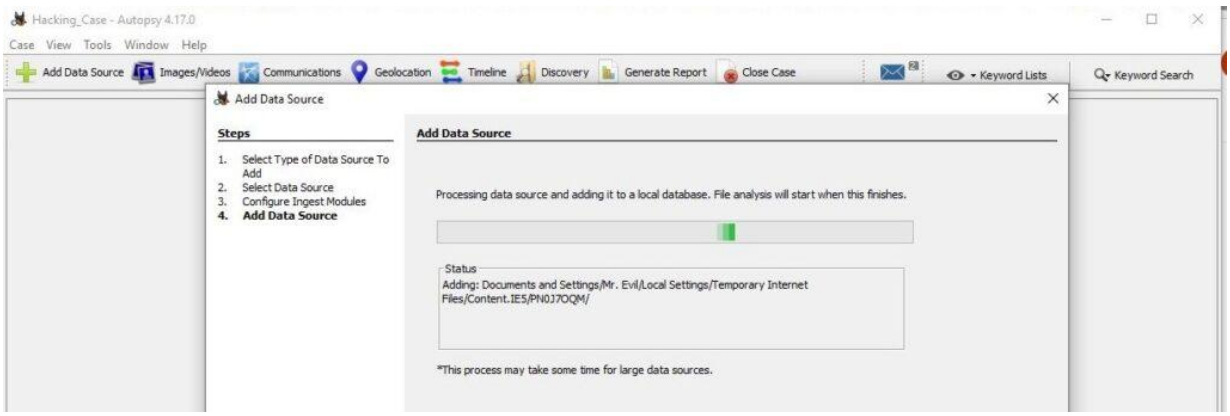


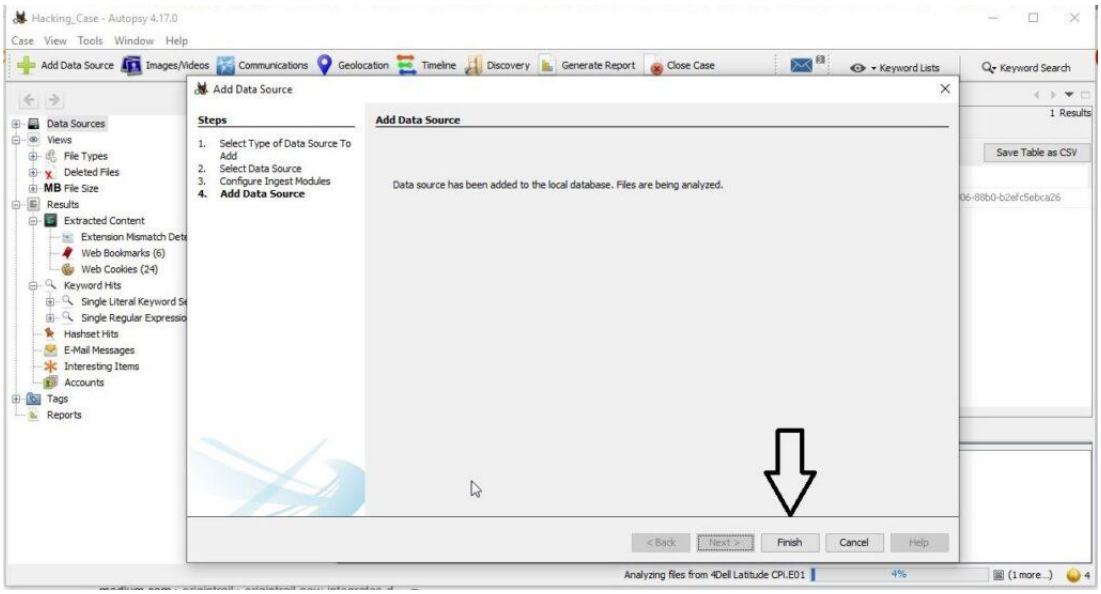
Select the Data Source. You need to download **two Encase Images**. Select the **first part** (**4Dell Latitude CPi.E01**)of the Encase images downloaded.



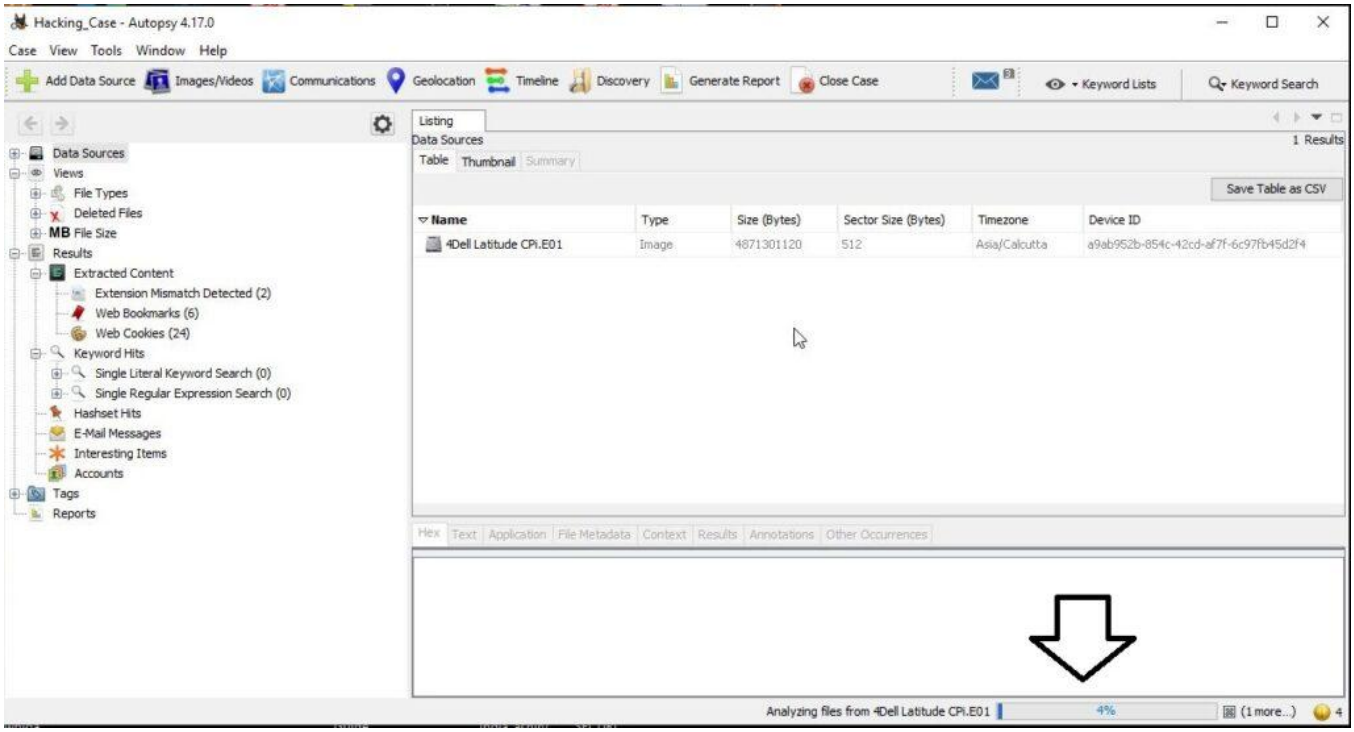
Next, select all the **ingest modules** you want to run. Ingest modules are all the tests that can be run on the image to gather information about it. These ingest modules include tests like **hash lookup, email parsing etc.** We **selected all** for this.



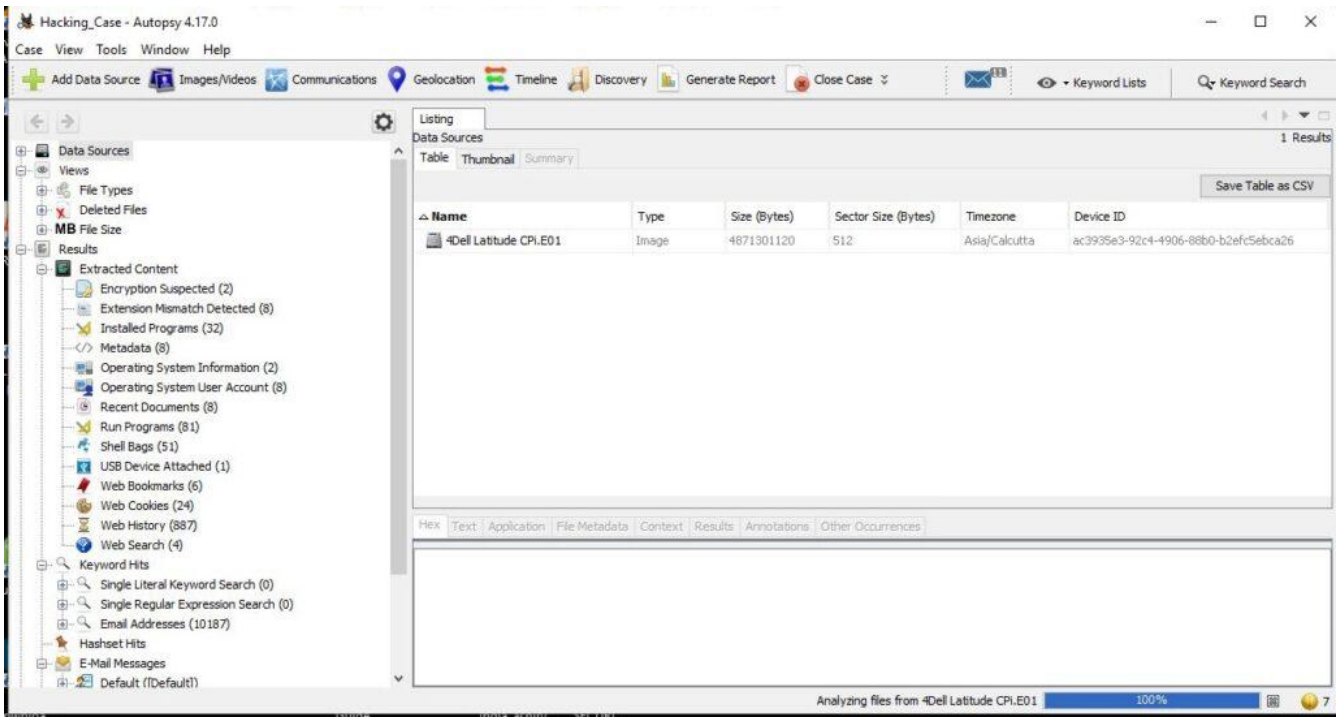




Autopsy will start analyzing the image. It may take some time to completely analyze the image. However, it will start **displaying findings** as soon as it finds them. Let the image analysis finish.



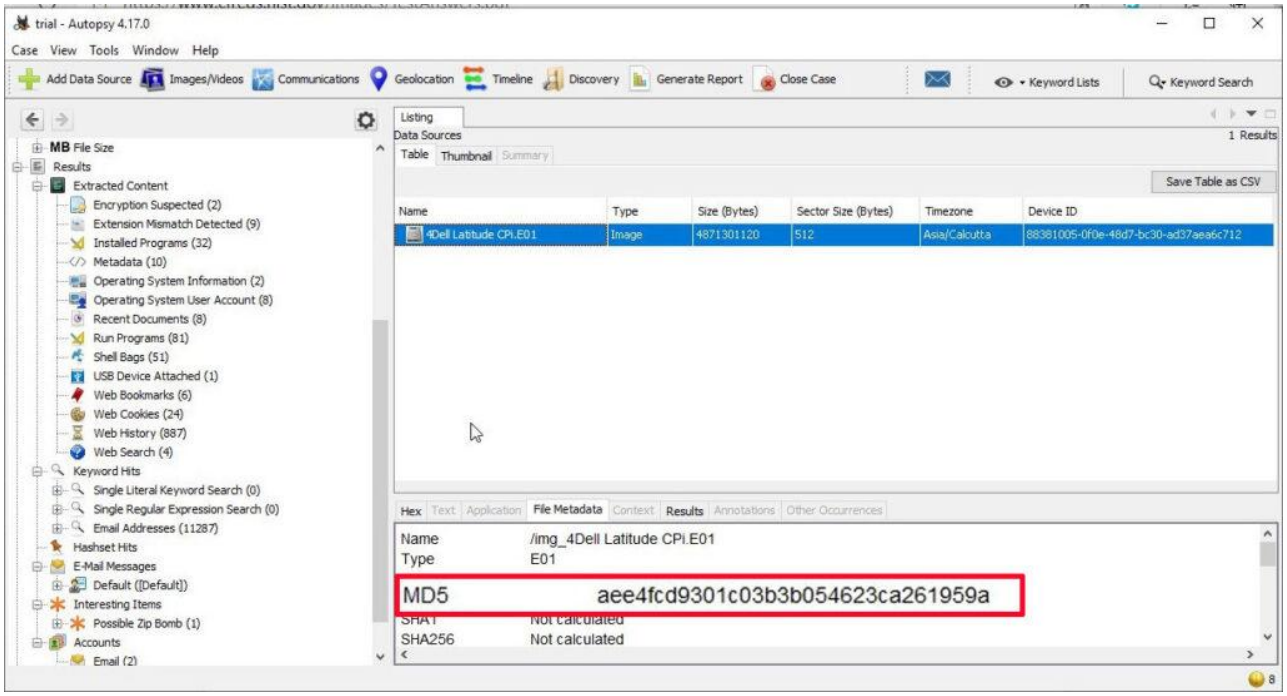
After the image analysis is finished, all the **extracted information** can be found on the left side of the program window.



Now, it's time to **start answering questions** related to the case.

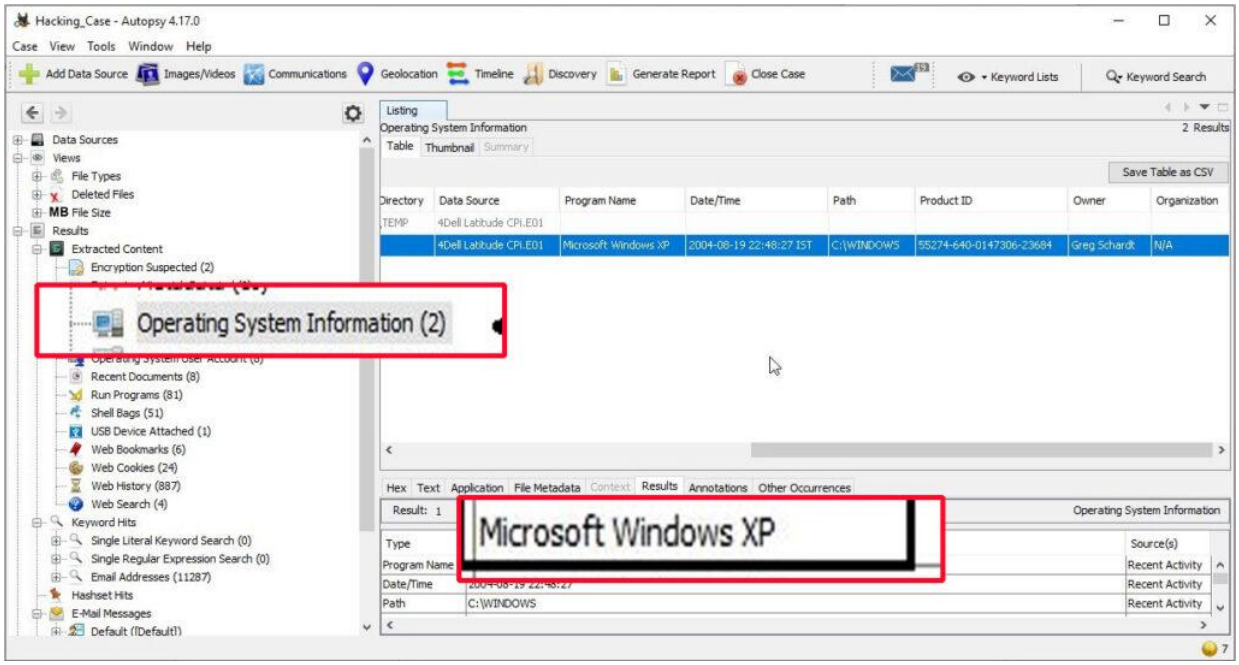
1. What is the image hash? Does the acquisition and verification hash match?

In Digital Forensics, as soon as a image is acquired to perform analysis on it, a hash is calculated to check if the file integrity is intact and not compromised. If the **acquisition and verification** hash do not match, it means our forensic analysis has changed the image which is not at all intended. The **image hash** is "AEE4FCD9301C03B3B054623CA261959A". It is found in the **File Meta data** section.



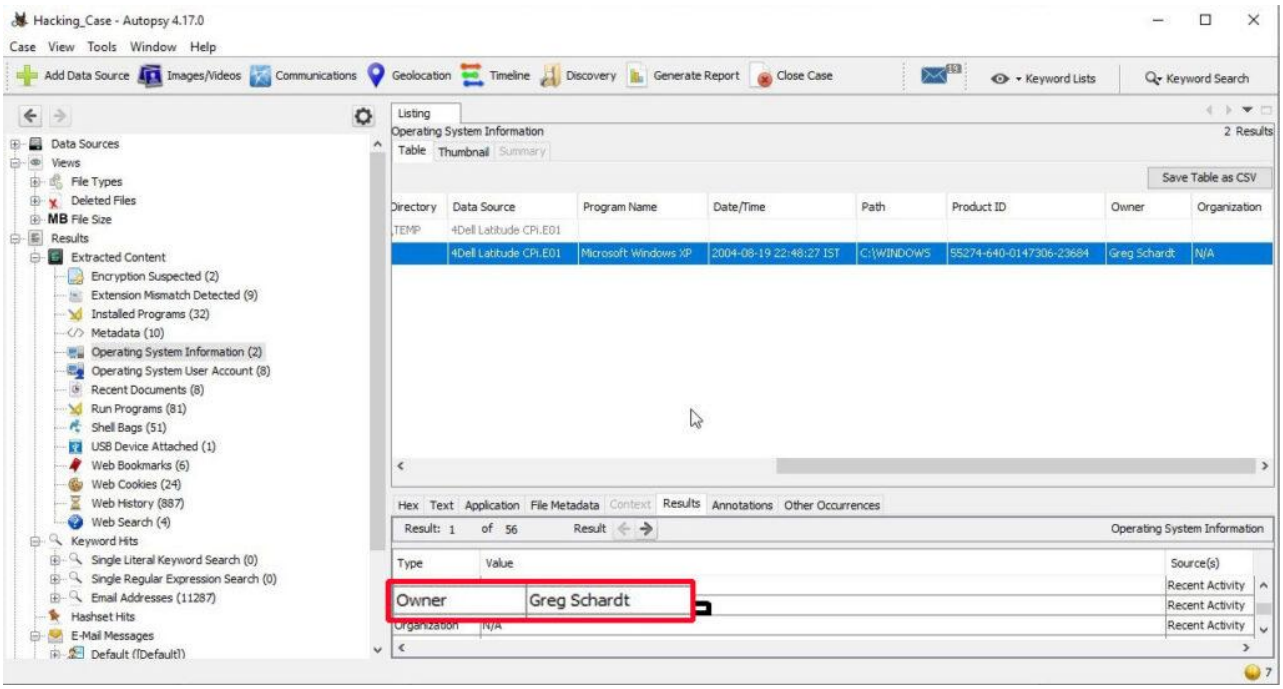
2. What operating system was used on the computer?

The operating system information can be found in the operating system information of the extracted content. The operating system is **Windows XP**.



3. Who is the registered owner?

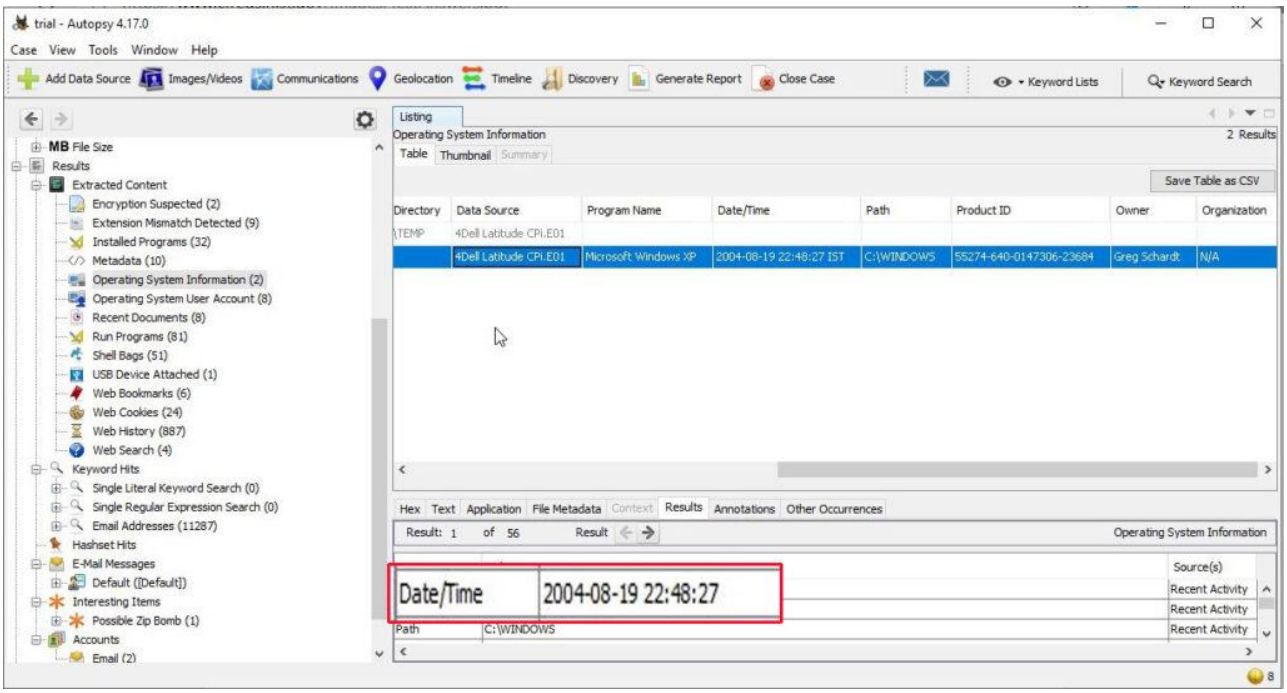
The information about the **registered owner of the computer** is found in the same operating system info section in extracted content.



The name of the owner of this computer is "**Greg Schardt**".

4. When was the install date?

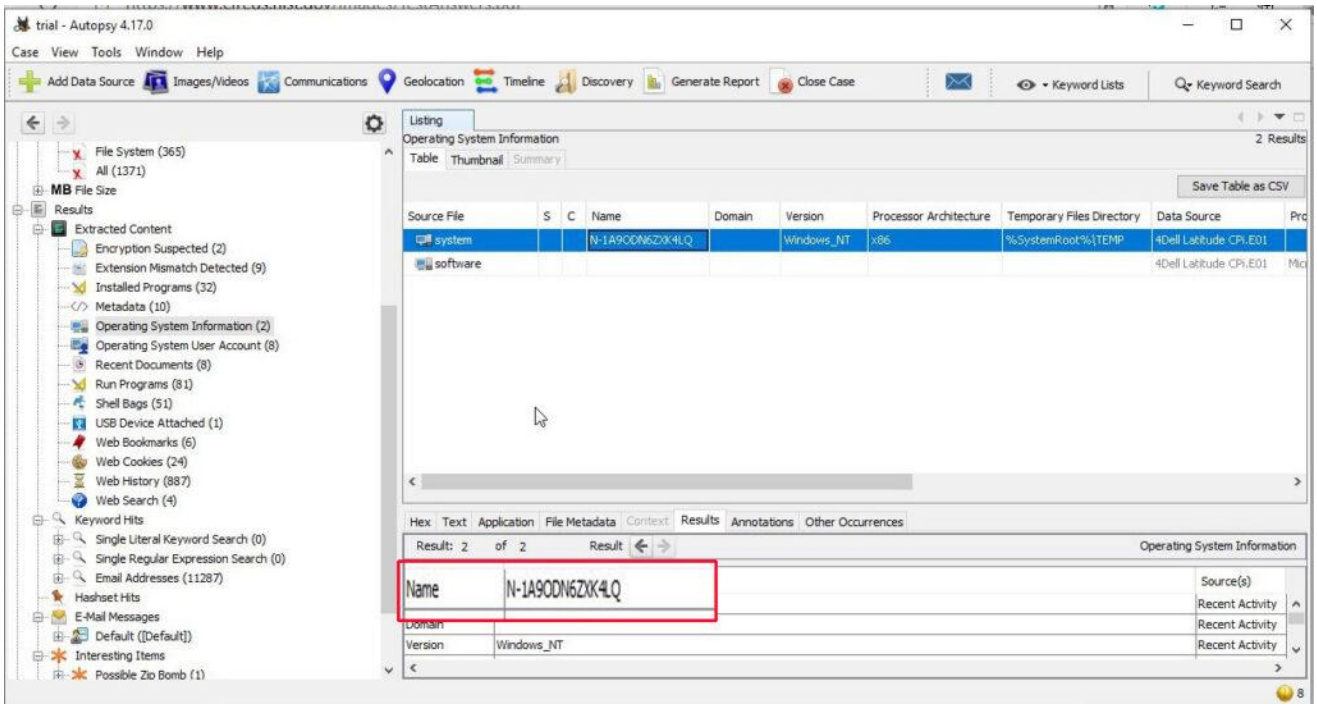
The **install date** can be found in the same operating system info section just below the OS information.



The OS on the computer was installed on **19-08-2004 22:48:27**.

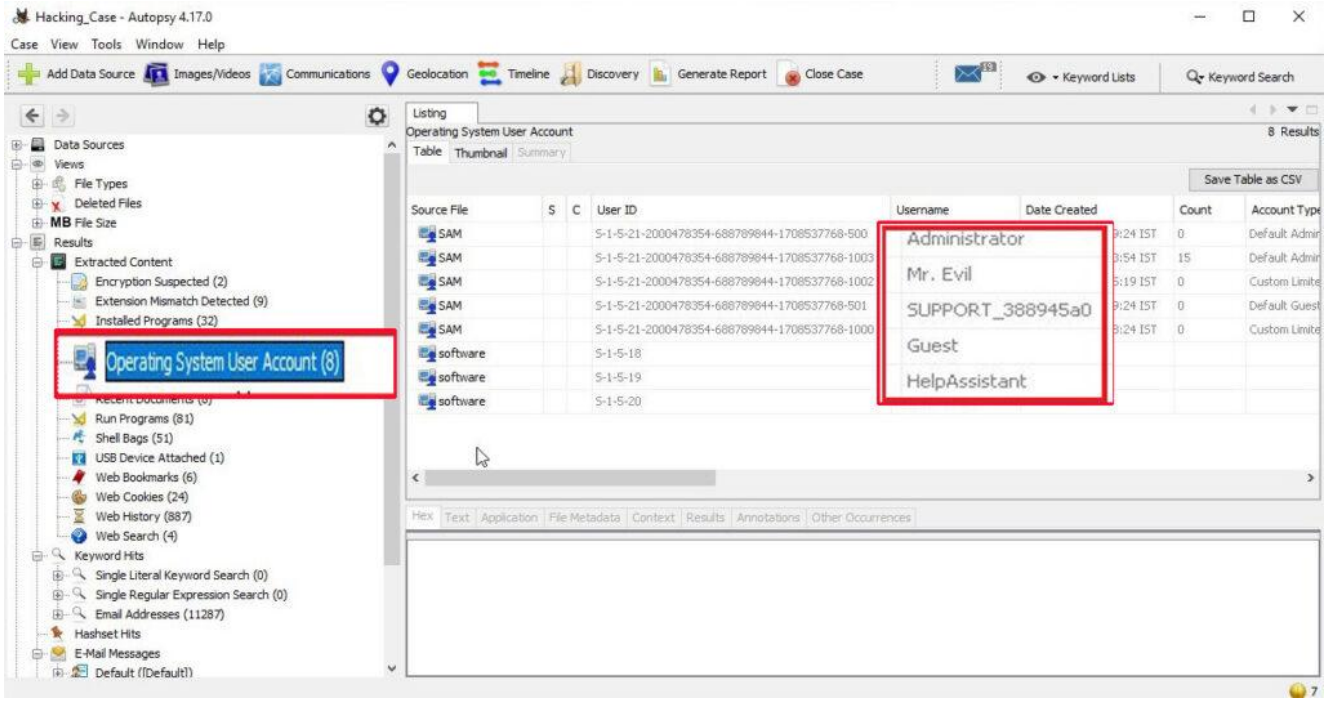
5. What is the computer account name?

The computer account name on this computer is found in the same section. The computer account name is N-**1A9ODN6ZXK4LQ**.



6. How many accounts are recorded?

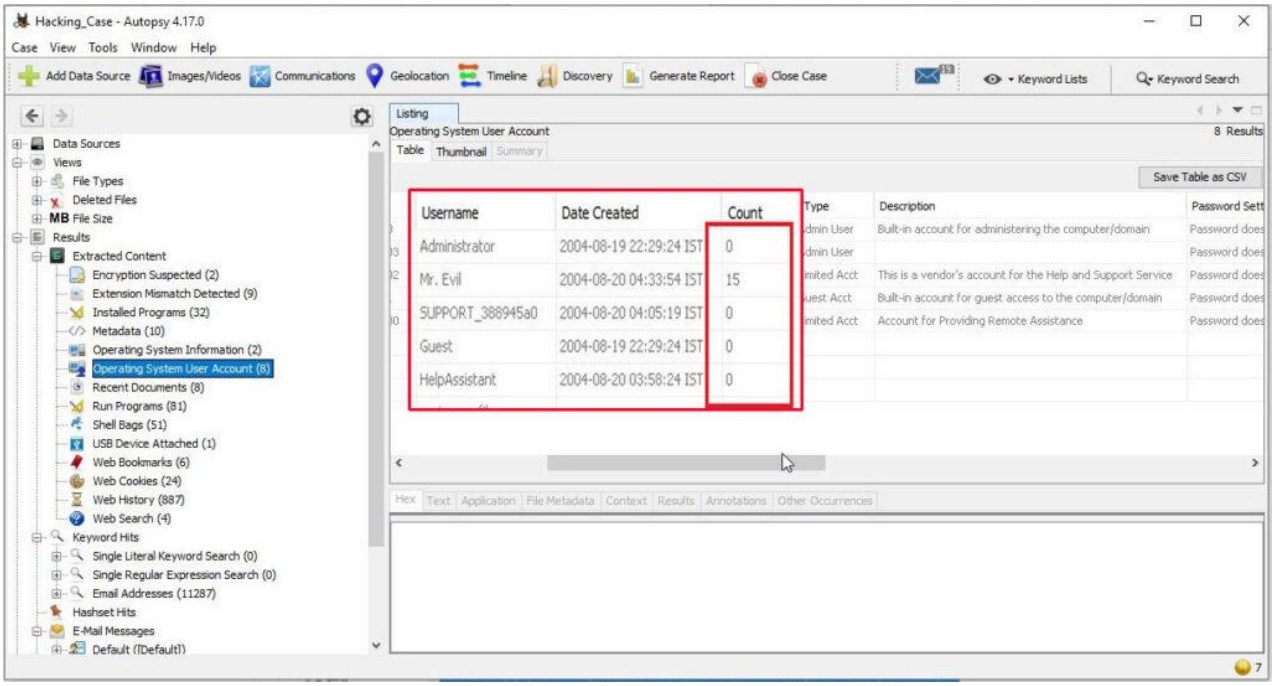
The information about the user accounts is found in the Operating system user account section.



There are total five user accounts on the target computer. They are Administrator, Mr. Evil, SUPPORT_388945a0, Guest and HelpAssistant.

7. What is the account name of the user who mostly uses the computer?

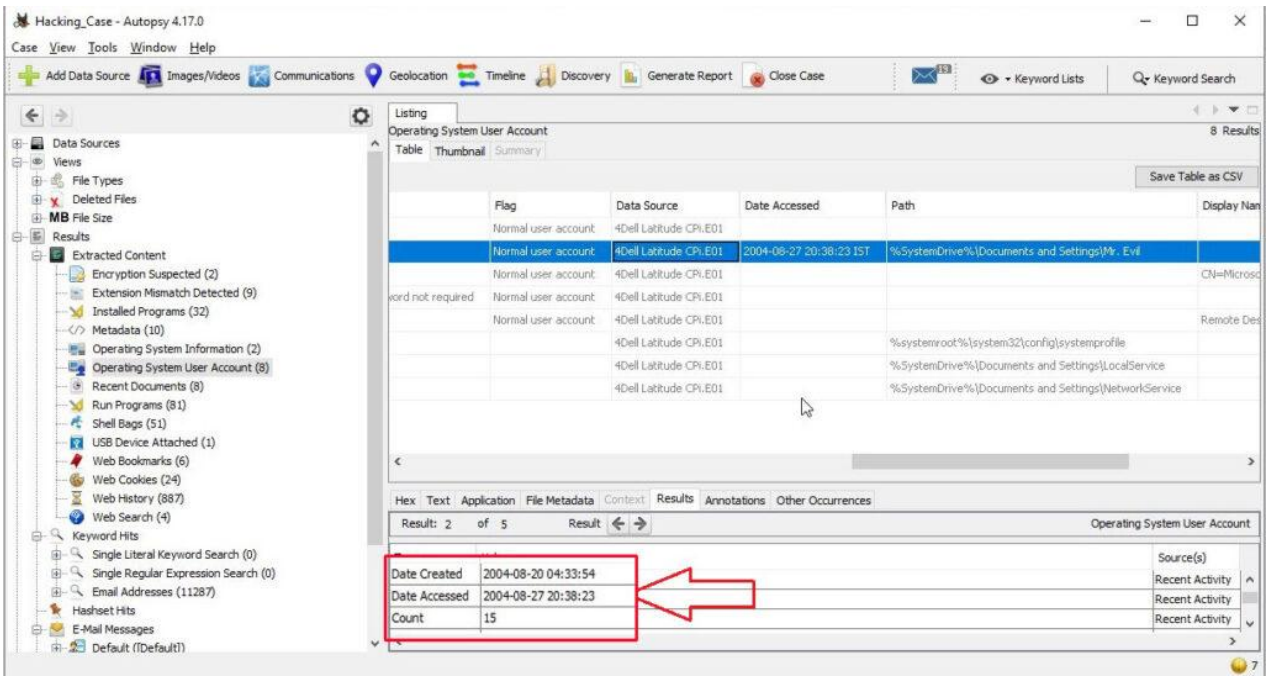
In the same section, the count section shows how many times the user logged in.



The user Mr. Evil has **logged in 15 times** while the others didn't even log in once. So Mr. Evil is the user who mostly uses the computer.

8. Who was the last user to logon to the computer?

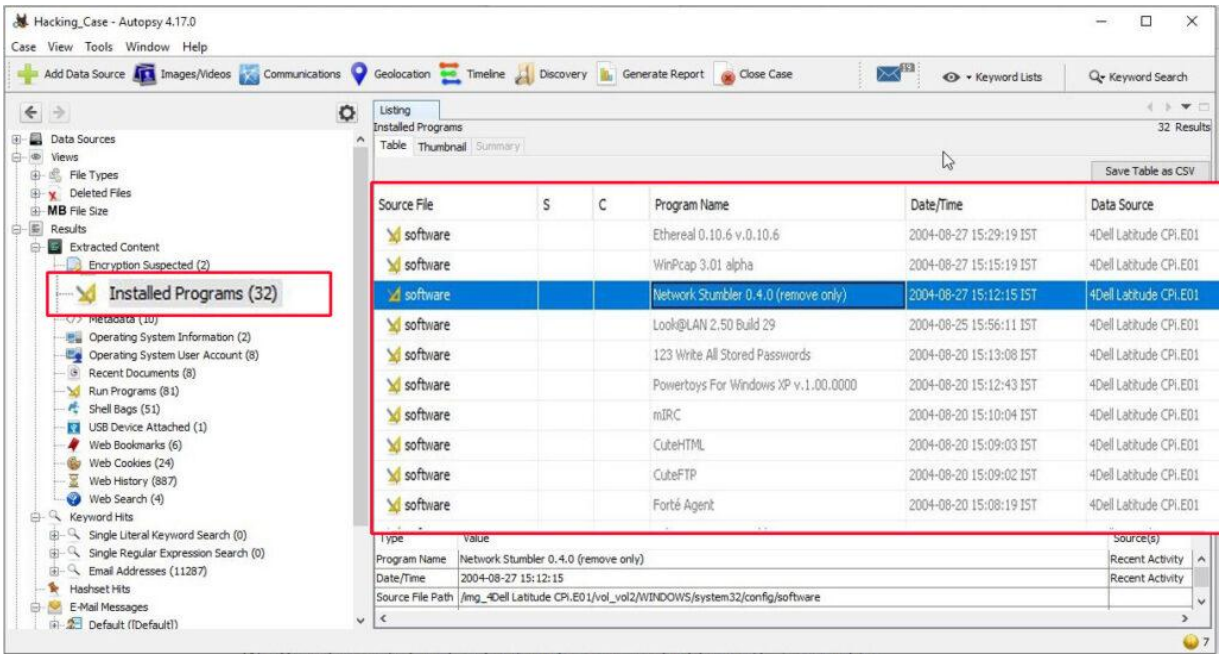
The information about the last user to logon to this computer can be found from the Date accessed column of the user account.



The last user to logon to this computer is Mr. Evil.

9. Find 6 installed programs that may be used for hacking?

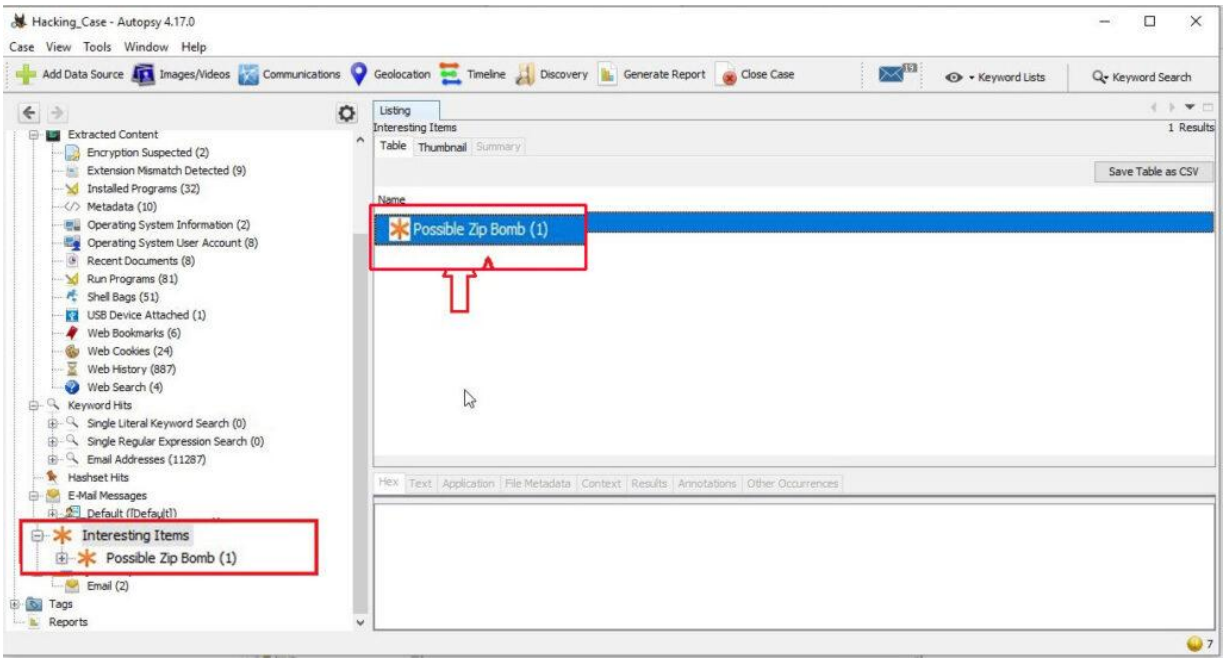
The programs installed on the computer system can be found out from the **Installed programs section** of the extracted content.



There are **total 32 programs** installed on the computer and from them, there are **seven programs** that can be used for hacking. They are Ethereal 0.10.6 v.0.10.6, Network Stumbler 0.4.0, Look@LAN 2.50 Build 29, 123 Write All Stored Passwords, CuteFTP, Cain & Abel v2.5 beta45 and Anonymizer Bar 2.0.

10. Perform a Anti-Virus check. Are there any viruses on the computer?

Malicious files (if any) are found in the Interesting Items section of the extracted content.

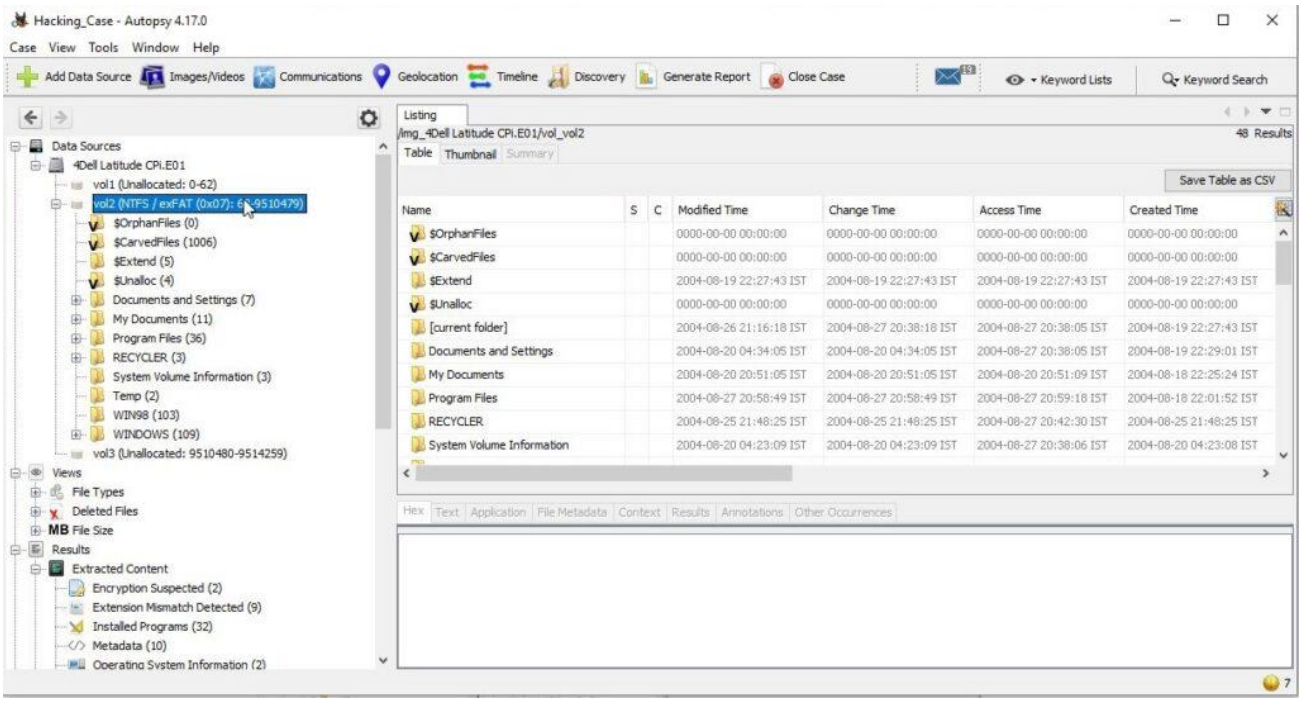


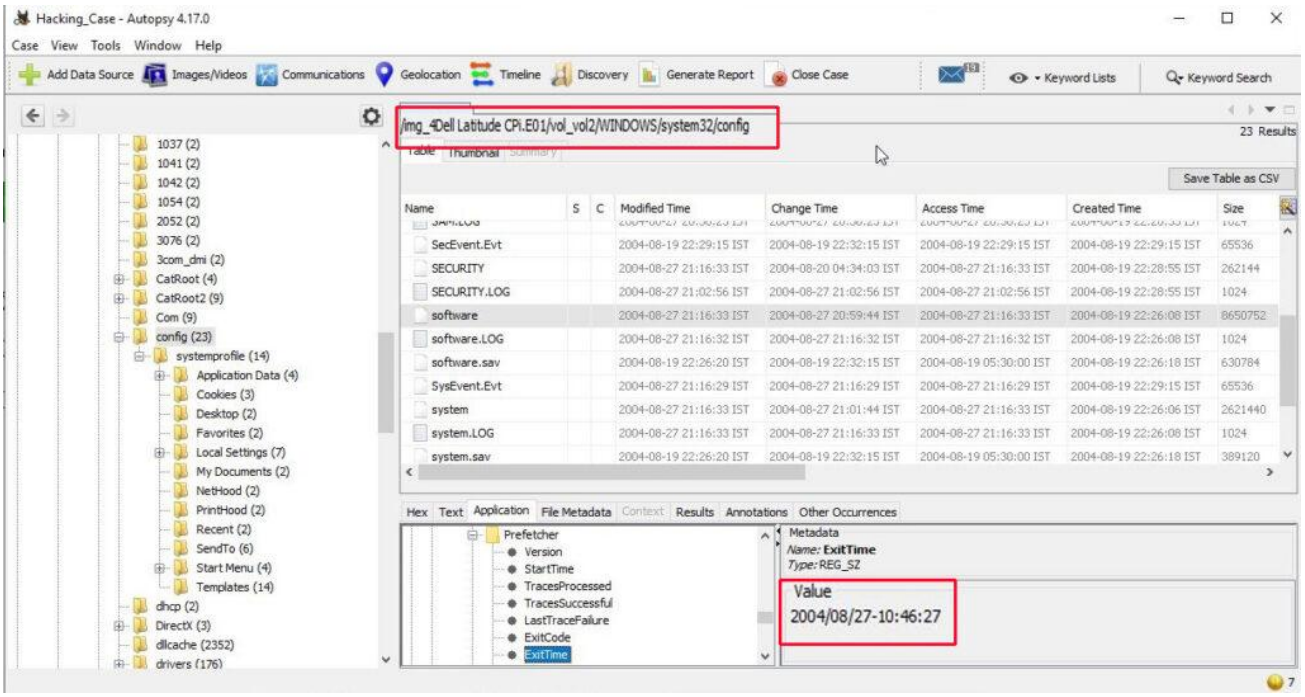
There is one malware present on the computer system. It is a zip bomb.

11. When was the last recorded computer shutdown date/time?

The last recorded shutdown date and time can be found out in the following file in Windows.

"C:\WINDOWS\system32\config\software\Microsoft\WindowNT\CurrentVersion\Prefetcher\ExitTime"

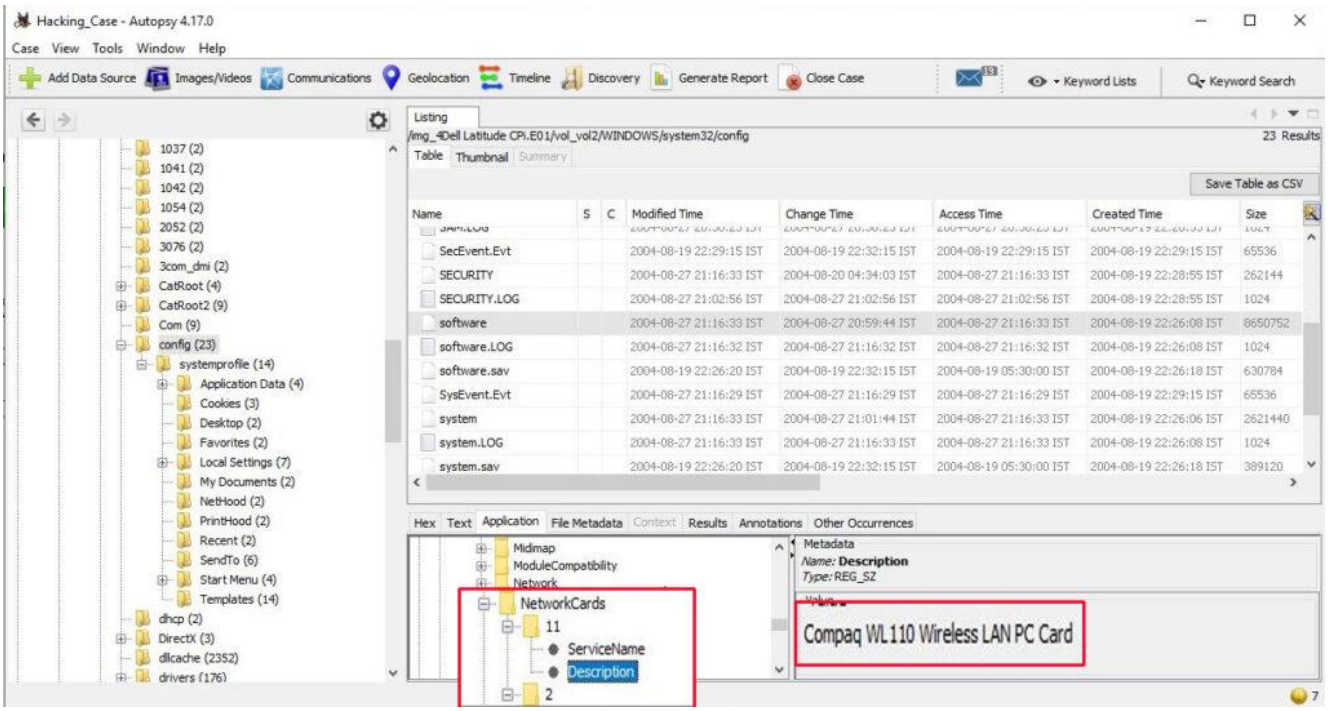




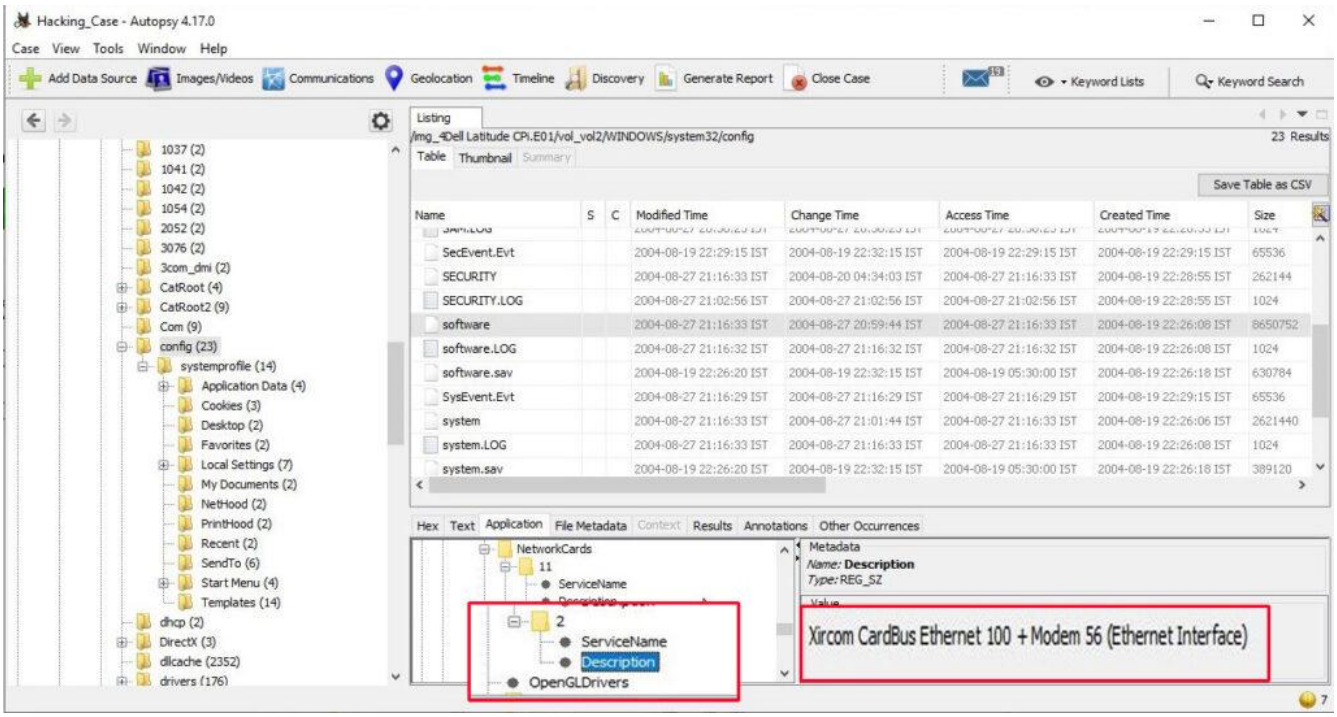
The shutdown date and time is 2004/08/27 10:46:27.

12. List the network cards used by this computer?

The information about the network cards on this computer can be found in the Windows file "C:\WINDOWS\system32\config\software\Microsoft\WindowNT\CurrentVersion\NetworkCards"

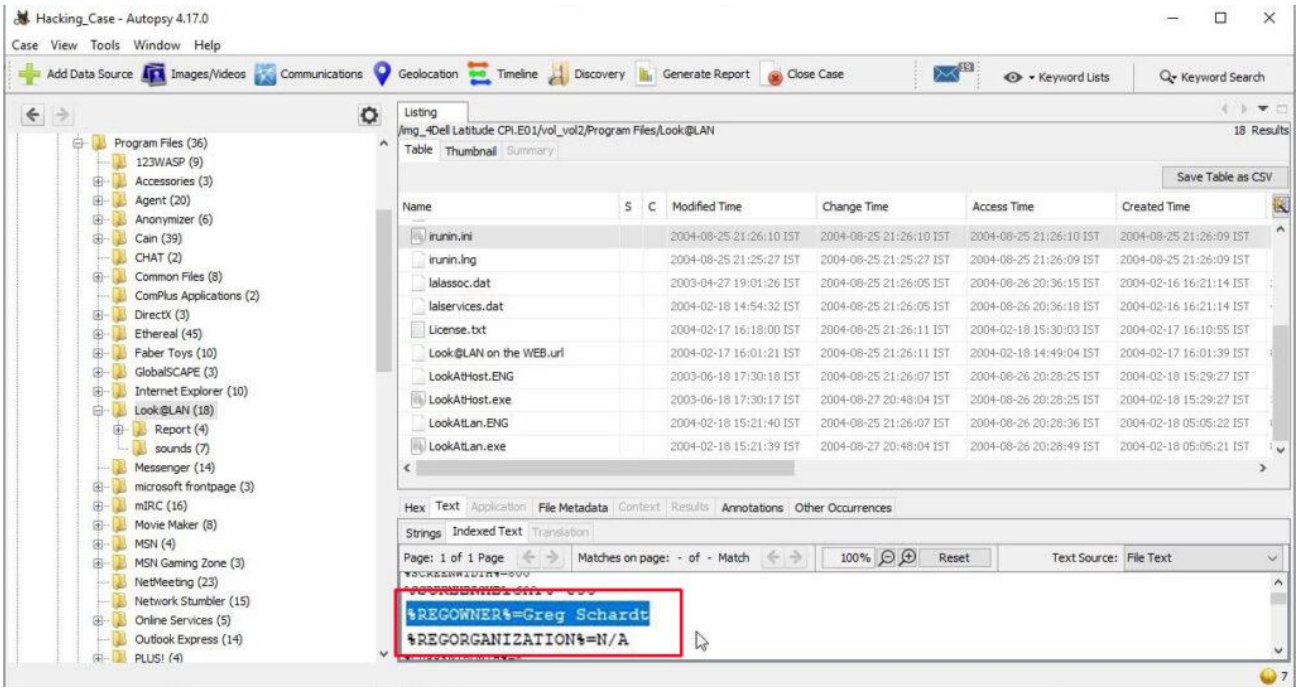


There are two network cards on this system. One is a Compaq WL 110 Wireless LAN PC Card and another is Xircom CardBus Ethernet 100 + Modem 56 (Ethernet Interface).



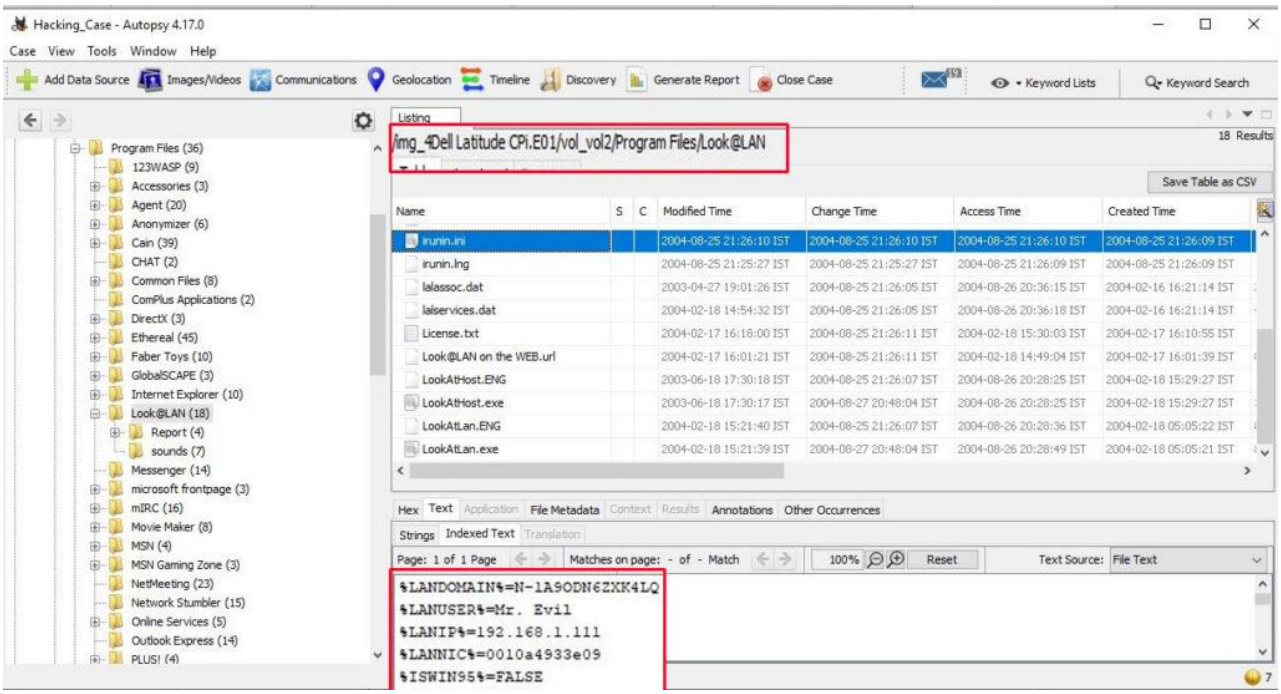
13. A search for the name of "G=r=e=g S=c=h=a=r=d=t" (The equal signs are just to prevent web crawlers from indexing this name; there are no equal signs in the image files) reveals multiple hits. One of these proves that G=r=e=g S=c=h=a=r=d=t is Mr. Evil and is also the administrator of this computer. What file is it? What software program does this file relate to?

The file that reveals all this information is **"C:\Program Files\Look@LAN\irunin.ini"**



His file belongs to the program **Look@LAN**.

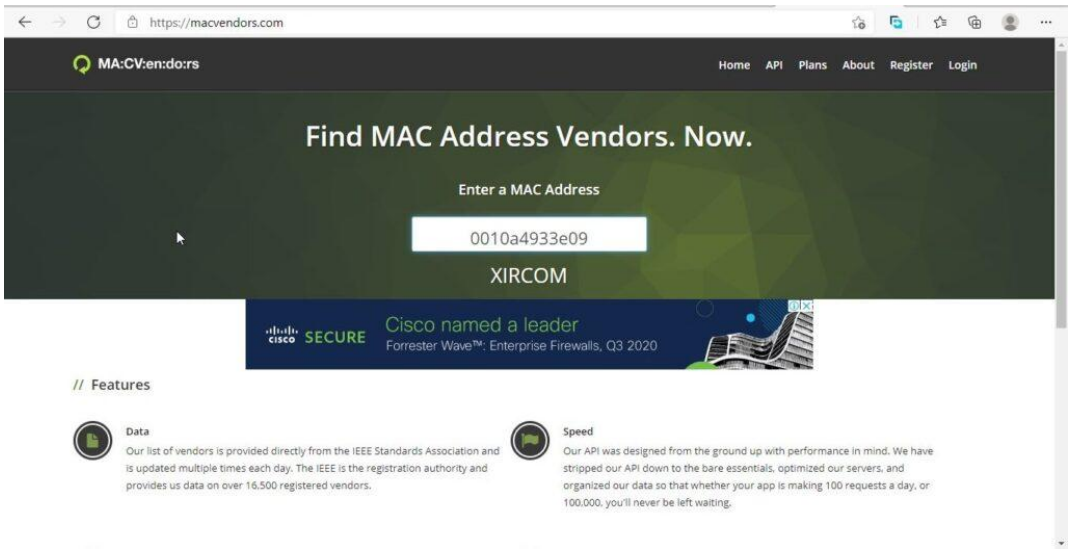
14. This same file reports the IP address and MAC address of the computer. What are they?



The IP address of this machine is 192.168.1.111 and the MAC address is 0010a4933e09. The LAN user is Mr. Evil. This confirms that Mr. Evil and Greg Schardt are one and the same.

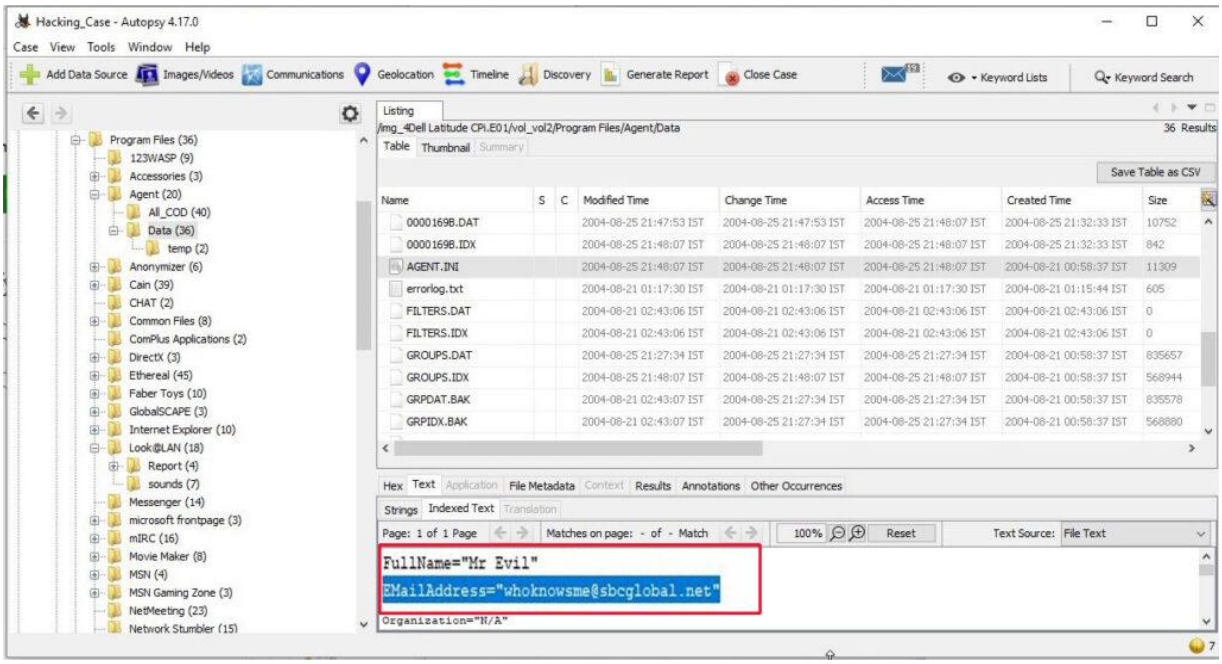
15. An internet search for vendor name/model of NIC cards by MAC address can be used to find out which network interface was used. In the above answer, the first 3 hex characters of the MAC address report the vendor of the card. Which NIC card was used during the installation and set-up for LOOK@LAN?

Media Access Control (MAC) address or the physical address is a 12 digit hexadecimal number hardcoded to the NIC card. The first 3 hexadecimal characters reveal the vendor of the NIC card. There are many websites which offer this service of knowing the vendor of the NIC card. Pasting the MAC address of the computer reveals the vendor. The Vendor of this NIC card is XIRCOM.



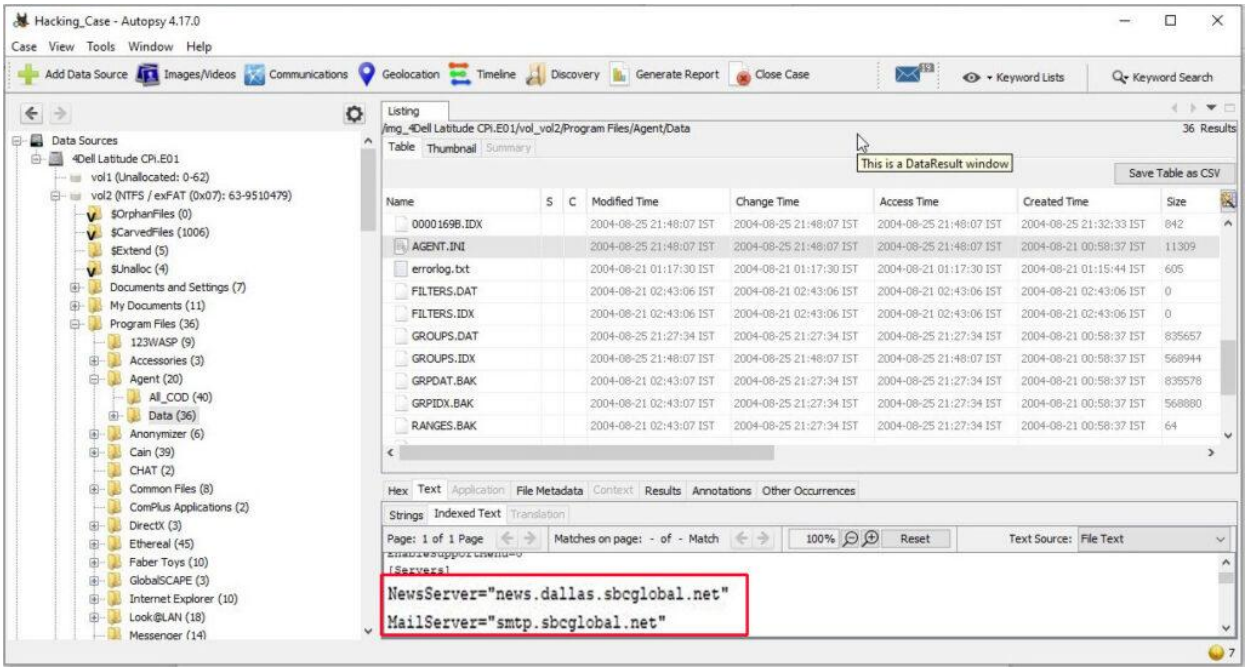
16. What is the SMTP email address for Mr. Evil?

SMTP or Simple Mail Transfer Protocol is a protocol used to send emails. The SMTP email address if present on the system can be found in "C:\Program Files \Agent\Data\AGENT.INI file". The SMTP email address is "whoknowsme@sbcglobal.net".



17. What are the NNTP (News Server) settings for Mr. Evil?

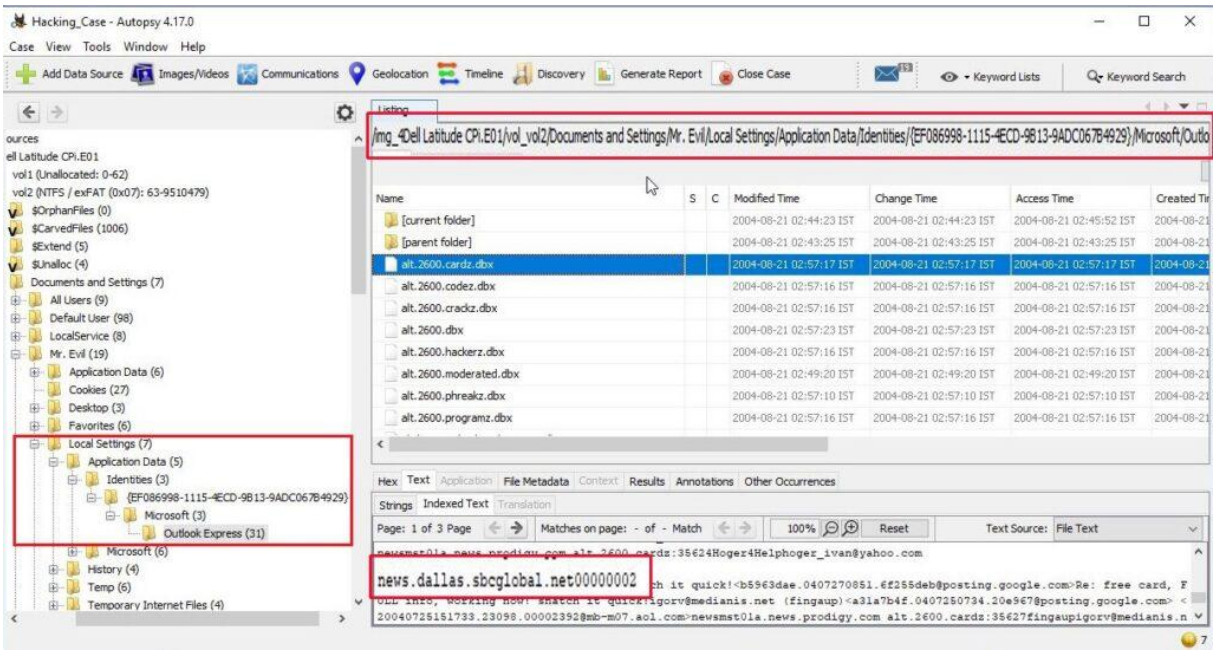
This information can be found in the same file as above.



The news server being used is “news.dallas.sbcglobal.net”.

18. What two installed programs show this information?

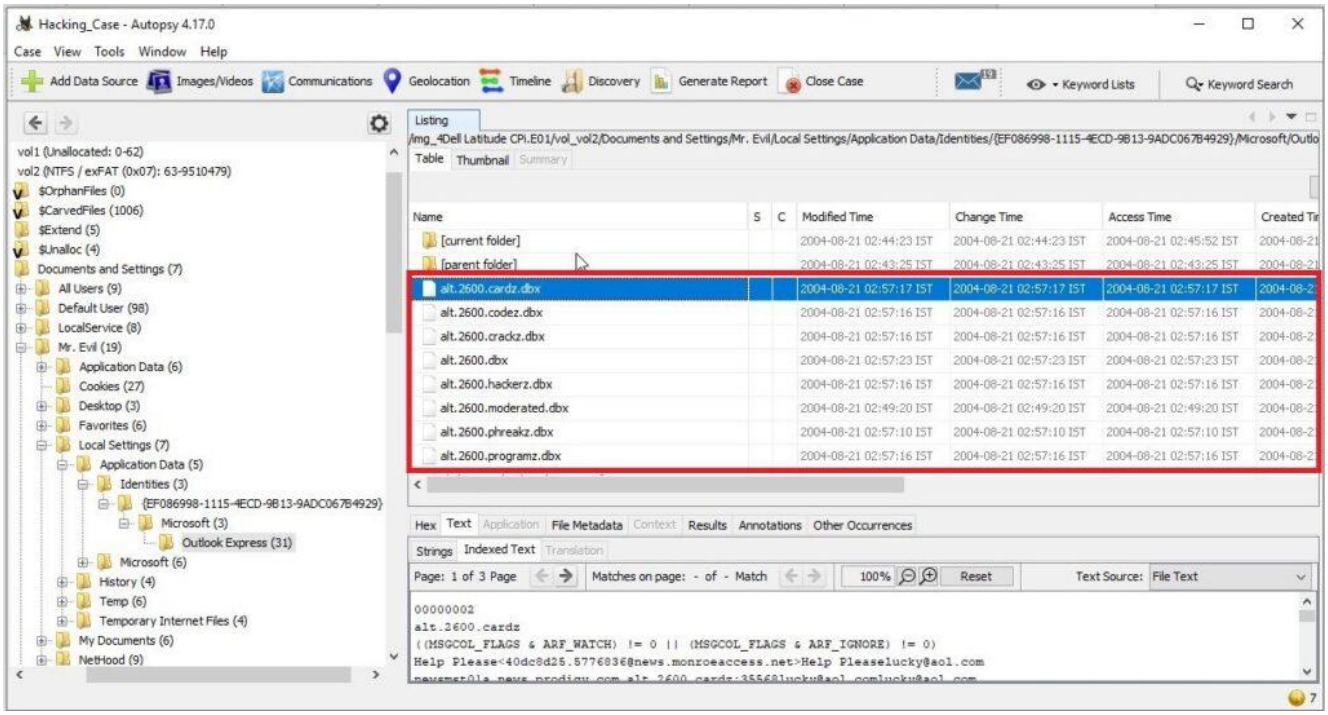
We searched for local settings of all programs and found the information about this news server in the local settings of Outlook Express.



We found this information in the documents and settings file (and above shown path) of user Mr. Evil.

19. List 5 newsgroups that Mr. Evil has subscribed to?

We can find this information in the same file as above.



User Mr. Evil subscribed to over 23 news groups. The news groups subscribed by the user Mr. Evil are,

1. Alt.2600.phreakz 2. Alt.2600 3. Alt.2600.cardz 4. Alt.2600codez 5. Alt.2600.crackz 6. Alt.2600.moderated 7. Alt.binaries.hacking.utilities 8. Alt.stupidity.hackers.malicious 9. Free.binaries.hackers.malicious 10. alt.nl.binaries.hack 11. Free.binaries.hacking.talentless.troll_haven 12. alt.hacking 13. free.binaries.hacking.beginner 14. alt.2600.programz 15. Free.binaries.hacking.talentless.troll-haven 16. alt.dss.hack 17. free.binaries.hacking.computers 18. free.binaries.hacking.utilities 19. alt.binaries.hacking.websites 20. alt.binaries.hacking.computers 21. alt.binaries.hacking.websites 22. alt.binaries.hacking.beginner 23. alt.2600.hackerz